

UC Berkeley, E29, Spring 2026

Lab 1: Reverse engineering, freehand sketching, and basic metrology practice

Plan to complete these tasks in the week of January 20–23
Deliverables are due on Gradescope by Monday February 2, 11:59pm

Objectives

This laboratory is a “warm-up” exercise to get you into the habit of analyzing manufactured products. We encourage you to be curious about which materials and manufacturing processes have been used to make the objects around you — and why.

We’ll also use this as an opportunity to practice freehand sketching, to illustrate the products you have analyzed.

Finally, you will get in some early practice in using handheld measurement tools for quantifying dimensional variability of manufactured objects. The measurements you collect today will be pooled with those of other students and you’ll analyze them in Lab 8.

Part 1: Choose two objects to analyze

Select two manufactured objects to analyze. You may have brought these in from home today; alternatively, there are plenty of objects available in the lab to choose from (please return them at the end of the lab). Some ideas for items to analyze include the following, but feel free to be imaginative when making your choices:

- Desk lamp
- Chair
- Stapler
- Electrical plugs
- Computer mouse
- Beverage container

Of particular interest, but not essential, would be if the two objects you select are designed to carry out a somewhat similar function, but have been made with different materials or processes. For example, a coffee mug designed to be used in the home might be made from molded clay fired with a ceramic glaze, whereas a portable mug for use on-the-go might be made from deep-drawn stainless steel connected to injection-molded components made from a plastic such as ABS (more on these processes later in the course). You can probably think of several reasons why different use-cases for products may motivate different material choices, and therefore the manufacturing processes used.

Part 2: Form hypotheses about materials and manufacturing processes

Look carefully at the objects you have selected and for each object fill out a copy of the “reverse engineering” table [available](#) in Word format on bCourses under Files > Labs > Lab 1. Each object will be made from one or more components; use one row of the table for each significant component. If the

object has many small components (like complex electronics, or many small screws), don't worry about covering all components exhaustively; focus on what you think are the most interesting and important components. It's fine if the object has only one component; in that case try to be as thorough and observant as you can in forming and describing your hypotheses.

You are not required to disassemble the objects, but you may wish to do so if it helps your analysis, if you own the object, if it is safe to do so, and if you are confident you can re-assemble it later!

Since we have barely started this class, we do not, of course, expect you to be an expert on manufacturing or to be able to identify the materials and processes accurately. It is possible to get full credit for this exercise if your hypotheses are logically explained, even if we happen to disagree with your final conclusion. Please just be as thorough as you can in "dissecting" the items in front of you, and, if you are curious, feel free to use online resources to find out how the items might have been made. There are many excellent online videos that describe how everyday consumer products are made.

Part 3: Make labeled sketches

Here, we ask you to make drawings that illustrate the objects you are analyzing. Use your judgement to make drawings that show the object as clearly as possible and label the drawings with the names of the components that you have used in your analysis tables.

We are deliberately not specifying anything else about how you make these drawings. You are free to choose every aspect of how you illustrate the objects you're analyzing. You may show one or more views of the objects, e.g. a single 3D representation or several 2D views taken from different angles. Also, we place no restriction on the media used. You can draw on paper and later scan¹ the drawings to upload, or sketch on your computer or tablet². Drawings can be monochrome line drawings or make use of shading or color as you see fit. The idea of this period of completely unconstrained sketching is to help you discover your own personal sketching style. In Lecture 3 we will start to talk about some specific techniques for sketching mechanical devices effectively.

Part 4: Optionally: analyze another pair of objects

If you have the time and inclination, you can repeat the exercise above for another pair of objects.

Part 5: Upload

Upload your reverse-engineering tables and labeled sketches to the Lab 1 assignment on Gradescope. You can upload PDF and image files.

Part 6: Metrology

This part of the lab is intended to give you some early practice of using hand-held metrology tools accurately, which may be useful during your production of parts for the printer. The practice will involve

¹ There are many apps available that use your phone camera for scanning. I personally like the "Scan Document" feature in the Dropbox mobile app (free), which, in its "whiteboard" mode, is particularly good for capturing crisp images of line drawings (including multicolored ones) and converting them to PDF or PNG files.

² e.g. Adobe Illustrator, available free via <https://software.berkeley.edu/adobe-creative-cloud>

collecting dimensional data for samples of manufactured objects, which will be pooled with other students' data and which you will analyze in Lab 8.

The following equipment is available today in the lab for your use:

- Electronic digital caliper
- Outside micrometer
- Small internal hole gage set

Your GSI will give a brief introduction of how to use the tools reliably and, in the case of the micrometer, how to read the Vernier scale.

In the lab are about 30 machined “widgets” (made by students in the Machine Shop orientation) and a set of bolts. The widgets have been uniquely numbered to enable pooling of the data gathered by multiple teams and therefore a broader analysis of the measurements.

The dimensioned drawing [here](#) shows the design of the widget and has five key dimensions labeled on it:

- A. Smaller diameter
- B. Distance from the machined flat to the opposite side of the component
- C. Overall length of component
- D. Inside diameter of the drilled hole that is closer to the machined flat
- E. Larger diameter (this surface has not been machined; it is as supplied in the bar of stock material)

Carry out the measurements detailed below and enter your data as you go into the pre-formatted [Excel spreadsheet available with this assignment](#).

Widgets. Make the following measurements for each one of *five* widgets randomly selected from those available in the lab:

- Dimension A, using the caliper;
- Dimension A, using the micrometer;
- Dimension D, using the caliper;
- Dimension D, using the hole gage plus micrometer.

Set the caliper to record in inches so that all measurements are taken in consistent units. Record each measurement to the full resolution of the measuring instrument used.

Bolts. Use the caliper to measure the lengths of five different bolts randomly selected from those in the lab.

Wax crayons. Use the caliper to measure the diameters of five different wax crayons randomly selected from those in the lab (measure the portion with the paper covering).

Felt-tip pens. Use the caliper to measure the diameters and lengths of five different felt-tip markers randomly selected from those in the lab.

Do not fill in the blue cells on the right of the spreadsheet yet – we will complete these in Lab 8.

Upload your spreadsheet with dimensional measurements to Gradescope (credit will be given for completion of this part, not for the measurements having any particular values). We will pool these data

and make them available (anonymously) to the whole class for analysis in Lab 8. Also keep the spreadsheet safe, ready for Lab 8.